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Paleomagnetism and rock magnetism of Fernando de Noronha, Brazil

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Seven sites from basaltic rocks of the Quixaba Formation (K/Ar age 3.4 ± 1.7 m.y.) on Fernando de Noronha (3.85°S, 32.4°W) yield a pole at 80.3°N, 234°E with $A_{95} = 2.9°$ and $K = 446$ after AF cleaning, and seven sites from the Remédios Formation (9.5 ± 1.1 m.y.) yield a pole at 84.5°N, 136°E with $A_{95} = 7.5°$ and $K = 65$. Combination of all published reliable data results in poles at 80.6°N, 216°E ($N = 21$, $A_{95} = 3.2°$, $K = 100$) and 86.0°N, 81°E ($N = 17$, $A_{95} = 4.9°$, $K = 53$) for the Quixaba and Remédios Formation, respectively (both with mixed polarity). The two pole positions differ significantly from each other. Seven sites from volcanics (K/Ar age 28.7 ± 2.5 m.y.) in the province of Fortaleza (3.8°S, 38.7°W) yield a pole at 66°N, 54°E with $A_{95} = 19.1°$ and $K = 10.9$. Rock magnetic investigations and chemical analyses on samples from Fernando de Noronha reveal as primary phase titanomagnetites with about 60% ulvöspinel in most cases. The titanomagnetites have undergone high-temperature oxidation and low-temperature oxidation of various degrees. The ferrimagnetic phases in the samples from the province of Fortaleza are magnetites with low-temperature alterations showing relative weak stability of remanence.

1. Introduction

The Atlantic island Fernando de Noronha (Fig. 1) is about 365 km northeast of Natal, Rio Grande do Norte, Brazil. The island constitutes the eastern end of a chain of volcanic seamounts. At the western end on the continent near Fortaleza (Ceará) volcanism is also found. There are similarities in the volcanic activities in both areas [1]. In this paper data on palaeomagnetism and rock magnetism of Tertiary volcanic rocks near Fortaleza and from Fernando de Noronha are presented. Previous palaeomagnetic investigations of Fernando de Noronha were carried out by Richardson and Watkins [2] and by Mazoires et al. [3].

Geology and petrology of the island have been described by Almeida [4], Gunn and Watkins [5] and Schwab and Bloch [6]. Two principal stratigraphic units are defined (Fig. 1): the older Remédios Formation consisting of phonolites, nephelinites, alkali trachytes, and alkali basalts, and the younger Quixaba Formation consisting of ankaramitic lavas. In addition there is a small outcrop (at site 59 in Fig. 1) which belongs to the São José Formation consisting of nephelinite basanite. K/Ar age determinations were carried out by Cordani [7]. Six samples from the Quixaba Formation yield ages between 1.81 and 6.64 m.y. with a mean of 3.37 ± 1.7 m.y., and 17 samples from the Remédios and the São José Formations ages between 8.02 and 12.32 m.y. with a mean of 9.47 ± 1.06 m.y. (The ages of the latter two formations do not differ significantly.)

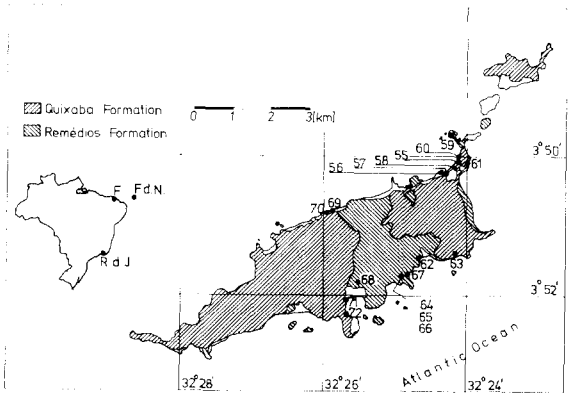


Fig. 1. Geological sketch map of Fernando de Noronha [4] showing sampling sites.

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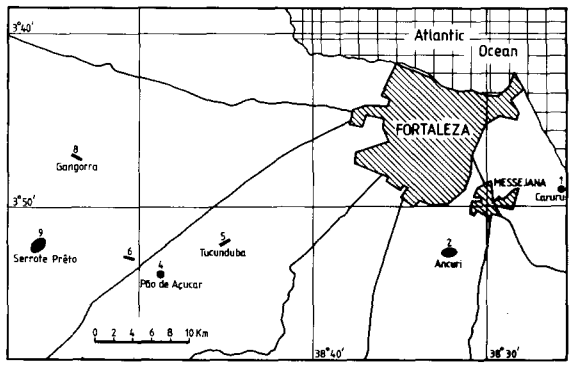


Fig. 2. Map showing sampling sites in the province of Fortaleza.

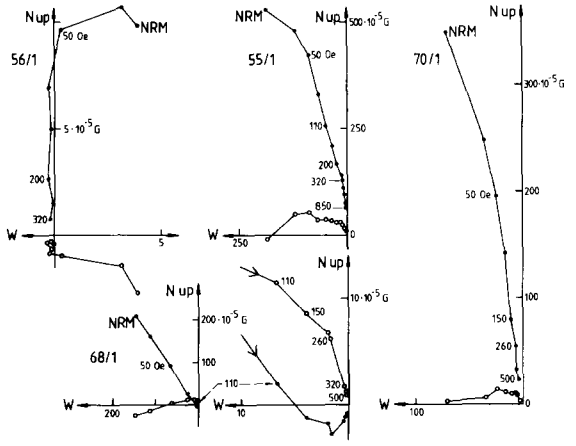


Fig. 3. Vector diagrams showing the variation of the remanence during progressive AF demagnetization of samples from Fernando de Noronha. Open and solid symbols indicate components in the vertical E-W and horizontal planes, respectively.

Tertiary volcanics near Fortaleza (Fig. 2) are composed predominantly of phonolites, trachytes and nepheline syenites which form necks, plugs, domes, and dikes [1,4,8]. Several authors [4,8] have pointed to relationships between these volcanics and those of Fernando do Noronha. Radiometric determinations with the Rb/Sr method yielded an age of 34 ± 2 m.y. and 28.7 ± 2.5 m.y. (five samples) with the K/Ar method [1,8].

2. Sampling and palaeomagnetic measurements

Sampling sites are shown in Figs. 1 and 2. From each site 5–13 samples (2.5 cm in diameter) were cored with a portable drill and specimens (2.3 cm long) were cut from the cores. The natural remanent magnetization (NRM) was measured with a Digico spinner magnetometer. Then pilot samples of each site were subjected to AF cleaning up to 1000 Oe (1 Oe = 79.58 A m^{-1}) peak value. In Figs. 3 and 4 vector diagrams show typical examples for the removal of viscous magnetization for sites from Fernando de Noronha and Fortaleza respectively. All other samples were subjected to AF demagnetization with appropriate peak values according to these results and the characteristic remanent magnetization (CARM) was measured.

The results for Fernando de Noronha are summarized in Table 1. For the Remédios Formation two sites with a VGP less than 50° latitude were rejected from the overall mean. Some neighboring sites of the Quixaba Formation were probably taken from the same flow because the directions are not significantly different on the 95% probability level applying the *F*-test. In this case the sites were considered as one site (see Table 1). Giving unit weight to each site the mean directions and pole positions of both formations are significantly (95% probability) different from each other. The pole of the Quixaba Formation is also different from the present geographic pole. The mean directions of both formations are different from present field or dipole field directions.

A combination of all available data (this paper, [2,3]) may be useful because site locations are then distributed more uniformly over the island. For the Quixaba Formation sites 1, 2, 3, 4, 5 and 7 from Richardson and Watkins [2] were used (some mistakes of their calculated pole positions have

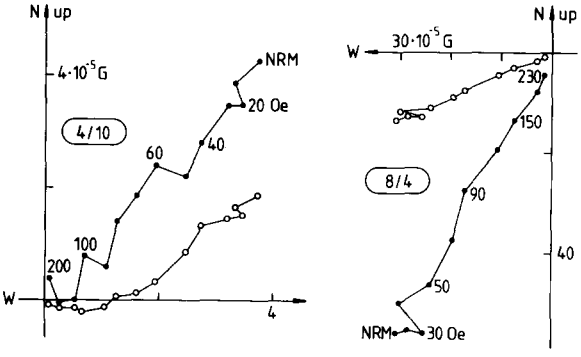


Fig. 4. Vector diagrams for samples from the province of Fortaleza. See Fig. 3.

TABLE 1

Site-mean palaeomagnetic results for Fernando de Noronha (ca. 3.85°S, 32.4°W)

A. Quixaba Formation

Site	NRM					MDF (Oe)	AF (Oe)	CARM					VGP	
	N	D	I	α_{95}	k			N	D	I	α_{95}	k	°N	°E
55, 61	9	342.3	−9.0	6.7	60	110, 80	300, 250	13	344.8	−11.6	4.9	73	74.7	229
57	9	345.1	−6.3	3.8	183	200	200	9	350.5	−7.9	2.2	528	80.5	236
58	7	348.2	−5.0	2.1	841	320	300	7	350.1	−3.8	2.3	634	79.9	248
60	4	342.1	+3.1	7.1	167	80	250	5	348.8	−4.9	5.4	199	78.7	244
63	7	351.1	−9.2	6.8	78	80	130	7	356.2	−12.3	5.2	134	85.6	205
69, 70	10	349.9	−5.8	3.3	222	200, 70	300, 150	11	349.6	−8.6	1.9	580	79.7	235
71	9	350.8	−8.6	3.4	222	300	300	9	352.3	−10.7	2.8	330	82.1	226
Mean of site means								7	350.3	−8.7	3.5	294	80.3	234
													$A_{95} = 2.9°$	
													$K = 446$	
Mean of site means, all data (this paper plus [2] and [3], see text; 12 normal, 9 reversed)										$N = 21$			80.6	216
													$A_{95} = 3.2°$	
													$K = 100$	

B. Remédios and Sao José (site 59) Formation

Site	NRM					MDF (Oe)	AF (Oe)	CARM					VGP	
	N	D	I	α_{95}	k			N	D	I	α_{95}	k	°N	°E
56	13	352.6	−2.7	6.1	42	80	100	9	354.1	+4.8	5.3	97	81.4	284
59	10	357.1	−12.5	2.8	293	50	150	10	2.7	−18.3	2.6	341	83.9	122
62/1	6	355.4	−32.5	3.9	291	100	120	6	359.9	−37.7	2.4	803	72.8	148
62/7	6	355.0	−17.3	3.2	441	50	120	6	1.7	−21.9	7.0	93	82.3	135
64	3	4.8	+7.4	27.5	21			no consistent result						
65	4	175.5	+2.5	14.2	43	500	300	3	175.7	+9.3	1.7	5600	85.8	226
66	5	11.3	−10.9	9.7	62	160	200	5	14.8	−11.9	8.8	76	75.0	67
67	6	346.6	−19.0	6.7	101	80	120	6	358.6	−26.9	2.7	617	79.5	155
68	12	no consistent result				40	300	6	(183.0	−65.6)	10.8	24	(38.3	330)
72	5	74.6	+46.8	23.8	11	20	120	6	(187.2	+65.0)	9.1	55	(46.4	141)
Mean of site means								7	1.1	−17.4	11.4	29	84.5	136
													$A_{95} = 7.5°$	
													$K = 65$	
Mean of site means, all data (this paper plus [2], see text; 13 normal, 4 reversed)										$N = 17$			86.0	81
													$A_{95} = 4.9°$	
													$K = 53$	

N = number of samples, D = declination, I = inclination, α_{95} and A_{95} = radius of 95% confidence circle, k and K = precision parameter, MDF = medium destructive peak field necessary to erase half of NRM intensity by alternating field demagnetization, AF = peak value of alternating field used. VGPs with least than 50° latitude and the respective directions were rejected from the mean.

been corrected). These sites are different from the sites of this paper. From Mazoires et al. [3] all data (8 sites) were used (their VGP data were taken from the graph). This yields, together with 7 sites from this paper, a mean pole position at 80.6°N, 216°E with $N = 21$, $A_{95} = 3.2°$, $K = 100$ (12 normal, 9 reversed). It is possible that two or three sites were considered twice for the mean. Rejecting the data from Mazoires et al. [3] yields practically the same pole: 80.5°N, 211°E with $N = 13$, $A_{95} = 3.2°$, $K = 168$.

For the Remédios Formation sites 10, 12, 14, 15, 18, 19, 24, 25, 27 and 28 [2] and the seven useful sites from this paper were used. All these

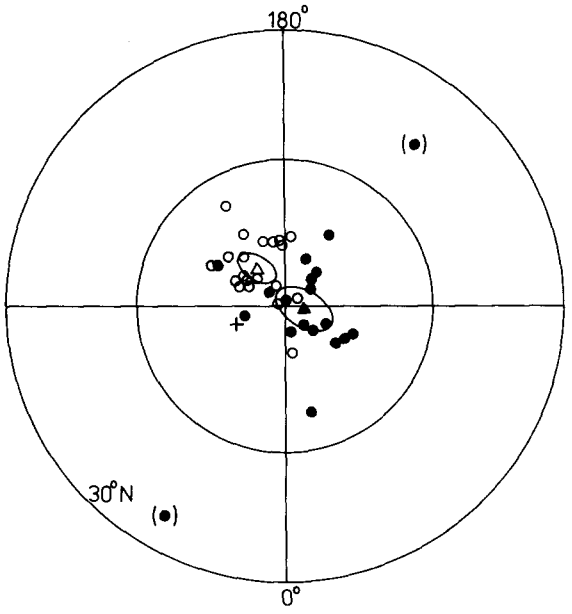


Fig. 5. Virtual geomagnetic poles of sites from Fernando de Noronha with the overall means (triangles) and circles of confidence. The cross denotes present dipole. A compilation from this paper and other publications [2,3], see text.

sites are probably different from each other. The mean pole position is at 86°N, 81°E with $N = 17$, $A_{95} = 4.9°$, $K = 53$ (13 normal, 4 reversed). The different mean pole positions for the two formations (see Fig. 5) were not recognized in the previous investigations [2,3].

The palaeomagnetic results for the province of

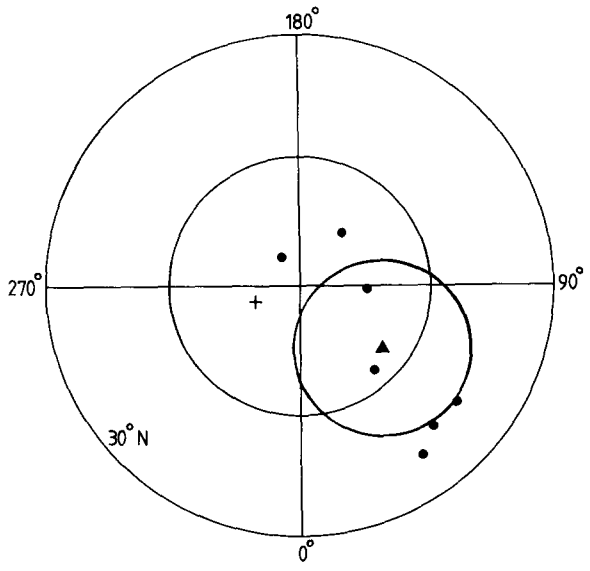


Fig. 6. Virtual geomagnetic poles of sites in the province of Fortaleza with the overall mean (triangle) and circle of confidence.

Fortaleza (Table 2) are of relatively poor quality. The within-site scatter is large and the stability of remanence is low (Table 2, Fig. 4). Also the between-site scatter is large. The overall mean yields a pole at 66°N, 54°E ($N = 7$, $A_{95} = 19.1°$, $K = 10.9$) with 4 normal and 3 reversed sites. This pole is different from the present dipole and it can be assumed that it is reliable (Fig. 6).

Tertiary palaeomagnetic results for South America are scarce [9–11]. They are similar to our results. It is interesting that the Early Pleistocene/Pliocene poles for South America have "left side" features [10,11].

TABLE 2

Site-mean palaeomagnetic results (CARM) for the province of Fortaleza (ca. 3.8°S, 38.7°W) [a]

Site	N	MDF (Oe)	D	I	α_{95}	k	°N	°E
1	19	70	45.6	−9.5	16.1	4	44.5	54
2	10	25	180.4	+32.6	9.5	22	76.1	140
4	9	60	25.4	+0.8	10.9	18	64.3	42
5	8	65	351.0	−13.2	6.7	55	80.6	213
6	6	55	224.8	−5.9	13.9	18	44.7	43
8	10	90	192.3	+24.5	8.2	32	74.8	89
9	5	85	46.8	+18.2	10.7	34	41.6	35
Mean	7		24.5	−8.4	22.4	8.2	66.0	54
							$A_{95} = 19.1°$	$K = 10.9$

[a] Legend see Table 1.

3. Rock magnetic studies

In order to identify the magnetic minerals several investigations were carried out including the determination of thermomagnetic curves and hysteresis parameters, X-ray studies, ore microscopy, and chemical analysis. The thermomagnetic curves $I_s(T)$ were measured in air on whole rock samples (moderately crushed) in a field of about 1800 Oe by means of a magnetic balance. The lattice constants were determined in a Debye-Scherrer camera with Co radiation on magnetically separated ore. The chemical analyses were carried out with a transmission electron microscope (TEM) with energy-dispersive equipment.

The observed $I_s(T)$ curves can be classified in three types as type a, b, and c (Fig. 7). Type a is due to a titanomagnetite or titanomaghemite with high titanium content. Type b has two Curie points, one originates from titanomaghemite and the other from magnetite with low titanium content. This is probably due to a low degree of high-temperature oxidation yielding exsolution of magnetite and ilmenite coexisting with unexsolved

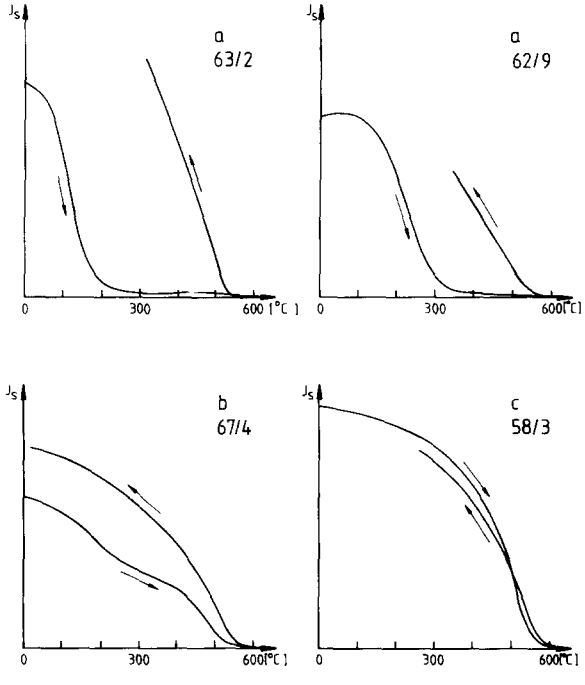


Fig. 7. Examples of thermomagnetic curve types a, b and c as defined in the text.

grains. Another possibility of origin are advanced low-temperature alterations resulting in a decomposition of titanomaghemite to magnetite and other phases. Type c, with a single Curie temperature around 500 °C, indicates magnetite as a result of high-temperature oxidation or as primary phase.

In Table 3 the results of the rock magnetic measurements are summarized and in Table 4 the chemical analyses. For all samples a small content of silicon was measured. Generally it is accepted that titanomagnetites and ilmenites contain practically no silicon. Therefore we assume that the silicon measured is due to a contamination by adjacent silicates. Omitting silicon and assuming stoichiometry the number of cations per formula unit was calculated (see Table 4). It is clear from the tables that the composition of the iron titanium oxides are relatively variable for the volcanics from Fernando de Noronha. The primary composition in the Quixaba and São José Formation was likely a titanomagnetite with x between 0.5 and 0.7. Due to local conditions the titanomagnetites were altered by high-temperature and low-temperature oxidation but remnants of the primary phase were found in all samples except 69/1. The disagreement between chemical analysis and rock magnetically derived titanium content in several cases is likely due to inhomogeneity of the samples (particularly for the Quixaba Formation). The primary composition in the Remédios Formation was in some cases a titanomagnetite with $x \approx 0.6$, in other cases magnetite with low titanium content (e.g. Sample 56 and 68). In both formations the variations of z indicate locally different low-temperature oxidation conditions. The site with the highest low-temperature oxidation [64] yielded no consistent palaeomagnetic result.

For the volcanics from the area of Fortaleza the Curie temperatures indicate a composition near magnetite in most cases (Table 3). For some samples no result could be obtained because of the very weak magnetization. The lattice constant determinations are only in two cases consistent with the Curie points, in other cases the Curie temperature and the lattice constant do not match on the contour diagram [13]. The rock magnetic results are in passable agreement with the chemical analyses yielding titanomagnetites with $x \approx 0.25$ (Table 4). In all cases the manganese content is relatively high. Ore microscopy showed no or very

TABLE 3

Rock magnetic data

Sample No.	a (Å)	T_1 (°C)	T_2 (°C)	Type of $I_s(T)$	x	z	Other phases	HTO degree	LTO
55/3 Q	8.452(5)	235	510	b	0.6	0.42	Ilm	I-II	+
57/1 Q	8.414(8)	535		c	0.13	0	Tmgh, Ilm	II	+ +
58/3 Q	8.411(3)	550		c	0.1	0	Tmgh, Ilm	II	+ +
60/4 Q	8.460(10)	240	515	b	0.52	0.15	Ilm	I-II	+
61/1 Q	8.474(7)	120	480	b	0.7	0.33	Ilm, Ht	II	+ +
63/2 Q	8.467(5)	165		a	0.65	0.33	Ht, Rt	I	+ +
69/1 Q	8.373(3)	570		c	0.18	0.77	Ilm, Ht	III	+ +
70/2 Q	8.448(4)	220		a	0.63	0.47		I	+ +
71/9 Q	8.411(5)	360	540	b	0.5	0.65	Ilm	I-II	+ +
59/1 SJ	8.484(2)	85		a	0.73	0.28	Ilm ?, Rt	I	+ +
56/8 R	8.415(2)	490		c	0.22	0.18		I	+ +
62/9 R	8.442(3)	300		a	0.5	0.35	Rt	I	+ +
64/3 R	8.365(4)	375		a	0.65	0.93	Ht, Rt ?	I	+ + +
65/5 R	8.379(2)	310	470	b	0.75(?)	0.9	Ht, Pbr	I	+ + +
66/1 R	8.412(3)	535		c	0.13	0	Tmgh, Ilm, Ht	I-II	+ +
67/4 R	8.452(5)	245	525	b	0.6	0.4	Ilm	II	+ +
68/1 R	8.412(3)	535		c	0.15	0.05	Ilm?	I	+
72/2 R	8.443(6)	315	505	b	0.45	0.25	Ilm	II	+ +
1/9 F	8.441(4)	525		c	?		Ht, Ilm	I-II	+ +
2/9 F	n.r.	200...400						I	
4/6 F	n.r.	505		c				I	+
5/2 F	8.479(10)	n.r.					Ht, Rt		
6/1 F	8.425(4)	490		c	0.2	0		I	+
8/4 F	8.488(4)	505		c	?				+ +
9/8 F	8.368(7)	565		c	0.2	0.8		I	+ +

Q: Quixaba Formation, R: Remédios Formation, SJ: São José Formation, F: volcanics from Fortaleza; a = lattice constant (errors in parentheses refer to the final digit); T_1 and T_2 = Curie temperatures (n.r. = no result); $I_s(T)$ curves are shown in Fig. 7; x = composition in $(1-x)\ Fe_3O_4 \cdot x\ Fe_2TiO_4$; z = oxidation parameter [12] and defines the extent of low-temperature oxidation (i.e. the fraction of the initial Fe^{2+} in the titanomagnetite converted to Fe^{3+}, $z = 0$ and $z = 1$ describe completely unoxidized and oxidized conditions, respectively), x and z were taken from the contour diagram for a and Curie temperature T [13]. Other phases (detected by ore microscopy, X-ray investigation or chemical analysis), Ht = hematite, Ilm = ilmenite, Pbr = pseudobrookite, Rt = rutile, Tmgh = titanomaghemite, HTO (high-temperature oxidation) degree after Wilson and Haggerty [14], LTO (low-temperature oxidation): + = weak, + + = moderate, + + + = strong (qualitatively estimated from the occurrence of shrinking gaps).

low high-temperature oxidation but different degrees of low-temperature oxidation. Very strongly altered are the samples from site 5: Only hematite could be detected by microscopy. However, the weak stability against AF treatment is not typical for this mineral (Table 2).

4. Hysteresis parameters

Relations between quotients H_{cr}/H_c (remanence coercitivity, coercitivity) and I_r/I_s (saturation remanence, saturation magnetization) can be used to distinguish different domain structure types [15]. The compilation in Fig. 8 shows that most samples from Fernando de Noronha have PSD domain structure. Samples from sites 56, 62, and 68 (all Remédios Formation) have MD structure or have a tendency towards MD structure. It is interesting that for sites 56 and 68 the primary phase is probably magnetite with low titanium content (Table 3). Generally the stability of the remanence for sites from the Remédios Formation is less than for those from the Quixaba Formation (Fig. 8, Table 1). This is reflected also in the frequency of the Q factors (NRM divided by induced magnetization in 0.3 Oe): ca. 35% of the samples from the Remédios Formation have Q factors between 0.1 and 1 which is rather low, and 65% between 1 and 10. For the Quixaba Formation all Q-factors are larger than 1. Particularly

TABLE 4

Chemical analyses of iron titanium oxides from Fernando de Noronha

Sample	*n*	Fe	Ti	Al	Mg	Mn	Si	Cr	Cations	Fe	Ti	Al	Mg	Mn	Cr	*x*
55/3 Q																
Tmt	8	88.7	9.9	0.4	0.1	0.6	0.4		3	2.669	0.298	0.012	0.003	0.018		0.6
Mgt?	2	96.7	0.6	0.8	0.2	0.2	0.8		3	2.945	0.018	0.024	0.006	0.006		
Ilm	4	45.0	47.4	1.1	4.2	1.1	1.0		2	0.911	0.960	0.022	0.085	0.022		
57/1 Q																
Tmt	12	80.2	17.2	0.5	0.5	0.8	0.6		3	2.425	0.520	0.015	0.015	0.024		
Mgt?	1	95.0	0.9	1.1	0.6	1.3	1.0		3	2.882	0.027	0.033	0.018	0.039		0.13
Ilm	1	50.9	45.2	0.8	0.5	1.0	0.8		2	1.035	0.919	0.016	0.010	0.020		
58/7 Q																
Tmt	5	79.2	16.4	0.8	1.5	1.2	0.8		3	2.398	0.496	0.024	0.045	0.036		
Mgt	1	94.0	1.9	1.0	1.6	0.7	0.8		3	2.843	0.057	0.030	0.048	0.021		0.1
Ilm	1	45.9	46.4	1.5	3.1	1.0	2.0		2	0.938	0.948	0.031	0.063	0.020		
60/4 Q																
Tmt	12	86.1	10.8	0.7	1.0	0.8	0.7		3	2.599	0.326	0.021	0.030	0.024		0.52
Ilm	1	47.9	48.0	0.1	2.6	1.1	0.3		2	0.961	0.963	0.002	0.052	0.022		
61/1 Q																
Tmt	6	74.0	21.8	0.8	0.9	1.1	1.4		3	2.252	0.663	0.024	0.027	0.033		0.7
Mgt	1	93.4	1.3	1.4	0.4	0.7	2.7		3	2.883	0.040	0.043	0.012	0.022		
63/2 Q																
Tmt	10	78.4	20.0	0.2	0.2	1.1	0.1		3	2.354	0.601	0.006	0.006	0.033		0.65
Mgt?	2	95.5	1.4	0.6	0.8	0.1	1.4		3	2.912	0.043	0.018	0.024	0.003		
Rt	1	3.8	91.5	2.1	0.3	0.1	2.1									
70/3 Q																
Tmt	8	76.4	18.0	1.4	1.6	0.8	1.8		3	2.334	0.550	0.043	0.049	0.024		0.63
71/2 Q																
Tmt	8	76.3	20.6	0.8	0.6	1.0	0.7		3	2.401	0.456	0.040	0.021	0.030		0.5
59/1 SJ																
Tmt	9	76.2	19.5	0.9	1.4	0.9	0.6	0.5	3	2.300	0.589	0.027	0.042	0.027	0.015	0.73
Rt	1	2.8	93.8	0.5	0.9	0.9	0.9									
56 R																
Tmt	8	87.4	8.0	0.4	0.3	3.3	0.4		3	2.638	0.241	0.012	0.009	0.100		0.22
62/9 R																
Tmt	8	78.5	16.8	1.3	1.7	1.1	0.6		3	2.369	0.507	0.039	0.051	0.033		0.5
Rt	1	10.3	85.7	1.0	0.8	1.0	1.1									
64/3 R																
Tmt	7	81.1	16.9	0.3	0.2	1.2	0.2		3	2.440	0.509	0.009	0.006	0.036		0.65
Ht?	1	95.9	2.5	0.5	0.2	0.5	0.5		2	1.926	0.05	0.010	0.004	0.010		
Rt?	1	16.5	79.5	1.1	0.9	0.7	1.4									
65/5 R																
Tmt	12	92.2	5.7	0.7	0.4	0.3	0.6		3	2.785	0.172	0.021	0.012	0.009		0.75
Ht?	1	94.0	0.5	0.3	1.0	0.3	3.9		2	1.956	0.010	0.006	0.021	0.006		
Pbr?	1	49.9	45.8	0.9	0.6	1.1	1.5									
66/1 R																
Tmt	12	87.2	10.0	0.6	0.4	0.6	1.2		3	2.648	0.304	0.018	0.012	0.018		
Tmt	1	90.5	4.8	1.3	0.6	0.5	1.8	0.5	3	2.765	0.147	0.040	0.018	0.015	0.015	0.13
Ilm	2	56.4	39.9	1.0	0.4	0.5	1.8		2	1.149	0.813	0.020	0.008	0.010		
67/5 R																
Tmt	9	80.5	16.3	0.9	0.9	1.0	0.4		3	2.425	0.491	0.027	0.027	0.030		0.6
Mgt	1	95.3	0.7	0.3	1.0	1.2	1.5		3	2.903	0.021	0.009	0.030	0.037		
Ilm	2	53.2	41.5	1.5	1.0	1.1	1.6		2	1.082	0.844	0.031	0.020	0.022		
68/3 R																
Tmt	8	88.2	6.4	0.6	0.2	4.0	0.6		3	2.662	0.193	0.018	0.006	0.121		0.12

TABLE 4 (continued)

Sample	n	Fe	Ti	Al	Mg	Mn	Si	Cr	Cations	Fe	Ti	Al	Mg	Mn	Cr	x
72/3 R																
Tmt	10	79.0	15.0	1.3	0.7	2.7	1.3		3	2.401	0.456	0.040	0.021	0.082		0.45
1/2 F																
Tmt	10	86.3	7.9	0.8	0.3	3.4	0.6	0.7	3	2.605	0.238	0.024	0.009	0.103	0.021	
4/2 F																
Tmt	5	90.0	6.0	0.3	0.1	3.4	0.2		3	2.705	0.180	0.009	0.003	0.102		
6/1 F																
Tmt	10	89.5	6.1	0.5	0.2	3.2	0.5		3	2.698	0.184	0.015	0.006	0.096		0.2
8 F																
Tmt	5	85.0	10.5	0.2	0.1	4.0	0.2		3	2.555	0.316	0.006	0.003	0.120		
9 F																
Tmt	6	83.1	11.1	0.8	0.3	3.7	1.0		3	2.518	0.336	0.024	0.009	0.112		0.2

Analyses are given in atom percent (total of cations 100%). Number of cations was calculated assuming stoichiometry and omitting Si. n = number of analyses. x from Table 3, to compare with Ti or the sum of Ti, Al, Mg, Mn. Mgt = magnetite, Tmt = titanomagnetite, Ilm = ilmenite, Ht = hematite, Rt = rutile, Pbr = pseudobrookite, see also Table 3. Question mark denotes uncertainty in mineralogy (e.g. instead of Mgt also Ht is possible).

samples from site 56 and 64 have small Q factors. For site 68 Q is very variable ($0.7 < Q < 180$) indicating magnetization by lightning. Sites 64 and 68 were not used for the overall mean direction (see Table 1).

For the Fortaleza area samples from site 1 show MD domain behaviour ($H_{cr}/H_c \gtrsim 4$, $I_r/I_s \approx 0.05$, $Q < 0.5$) consistent with a relatively unstable remanence (Table 2), samples from site 9 have PSD domain structure. The other samples could not be measured because of their weak magnetization.

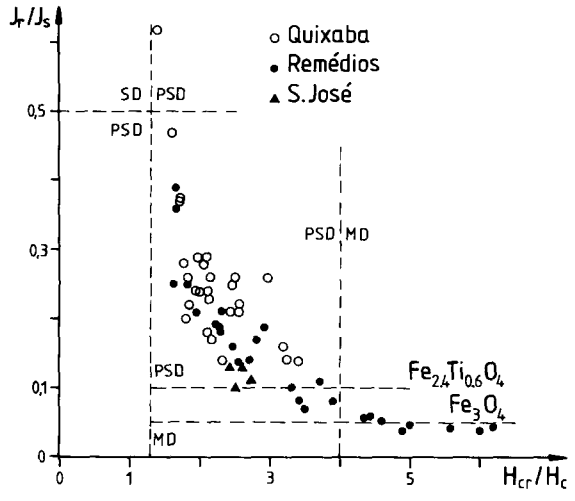


Fig. 8. I_r/I_s vs. H_{cr}/H_c for samples from Fernando de Noronha. The horizontal and vertical lines indicate the single-domain (SD), pseudo-single-domain (PSD) transition and PSD-multi-domain (MD) transition, respectively [15].

5. Conclusions

(1) The mean pole positions for Pliocene and Upper Miocene volcanic rocks from Fernando de Noronha (Quixaba Formation, Remédios and São José Formations, respectively) are different from each other. This was not recognized in previous studies.

(2) The ferrimagnetic minerals are titanomagnetites, many of which have undergone high-temperature oxidation. This implies subaerial eruption. Most samples have also undergone low-temperature oxidation of various degree.

(3) The initial titanomagnetites (before high-temperature oxidation) have in most cases a composition with $x \approx 0.6$, for the Remédios Formation in some cases with $x \approx 0.2$.

(4) The hysteresis parameters measured can be interpreted in terms of PSD structure for the Quixaba and São José Formation and PSD and MD structure for the Remédios Formation. The stability of remanence is moderate and somewhat weaker for the Remédios Formation. It seems certain that the CARM represents the ambient field when the rocks formed.

(5) In the volcanics of the Fortaleza area the ferrimagnetic minerals are titanomagnetites with

low titanium content ($x \approx 0.25$) in a certain degree of non-stoichiometry due to low-temperature oxidation.

(6) The remanence of the Fortaleza samples has weak stability. The result for the mean pole is poor but reliable.

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